

UNIT 1: Fundamental Concepts and Definitions

Syllabus topics: Scope of Mechanical Engineering; machines & mechanisms; classification of machines; thermodynamic systems & laws of thermodynamics; modes of heat transfer; materials classification & selection; manufacturing processes / mechanical working of metals.

Question 1

		Harcourt Butler Technical University Kanpur		MID TERM-I	
Branch	ME/CE/EE/CSE/IT/ET		Program	B.Tech	
Course Name	Introduction to Mechanical Engineering		Semester	2	
CourseCode	NME 102		Year	1	
Time: 1:00 Hr	Answer All Questions		Maximum Marks	15	
Session	2025-26				
Knowledge Level (KL)	K1:Remembering	K3:Applying	K5:Evaluating		
	K2:Understanding	K4:Analysing	K6:Creating		
Q. No	Questions		Marks	COs	KL
1a	Mention the statement of the first law and the second law of thermodynamics with examples.		2	CO1	K2

MID TERM-I | NME 102 | ME/CE/EE/CSE/IT/ET | 2025-26 | Q1(a) First & Second law of thermodynamics, Q1(b) Modes of heat transfer

Question 2

Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analyzing	K6: Creating	
Note: Attempt <u>all</u> questions. Each question carries the marks as shown against them.				
Q.No	Questions	Marks	COs	KL
	Describe various modes of heat transfer with applications.			

END SEM | NME-101 | Technology (BE/BS-MS/BT/CH/FT/LT/OT/PL/PT) | Q1(a) Modes of heat transfer, Q1(b) Manufacturing techniques

Question 3

Note: 1. Attempt ALL questions. Each question carries the marks as shown against them 2. Assume the missing data, if any.				
Q.no.	Questions	Marks	COs	KL
1a	Define Machine and Mechanism with example. Differentiate between machine and mechanism.	2	CO1	K1
1b	What are the three modes of heat transfer. How are they different from each other?	2	CO1	K2
1c	Give the classifications of materials based on their properties and characteristics.	2	CO1	K2
1d	What are the common manufacturing processes involved in the mechanical working of metals?	2	CO1	K2
1e	Classify machines on the basis of their functions and operation.	2	CO1	K2
2(a)	If the magnitude of the resultant force is 500 N, directed along the positive y axis, determine the magnitude of force F and its direction θ . [refer Figure-1].	5	CO2	K3
2(b)	Two blocks A and B of weight 100 N and 50 N are tied together by a rope parallel to the plane as shown in Figure 2. The coefficient of friction between block A and the plane is 0.2 and that between block B and the plane is 0.5. Determine the value of the	5	CO2	K3

END SEM | NME-102 | CE/EE/ET/ME/CSE/IT | 2022-23 | Q1(a)-(e) Machine vs Mechanism, heat transfer, materials, manufacturing, machine classification

Question 4

Q.No	Questions	Marks	COs	KL
1	(a) Define conduction, convection and radiation. (b) Differentiate between machine and mechanism. State first law and second law of thermodynamics.	5+5	CO1	K2, K1

End Semester | Q1(a) Conduction/convection/radiation, Q1(b) Machine vs mechanism + laws of thermodynamics

Question 5

NOTE: Maximum Marks:50, Attempt all Questions				
Q. No	Questions	Marks	COs	KL
1a	State the key differences between a mechanism and a machine.	2	CO1	K1, K2
1b	State the zeroth and the third law of thermodynamics	2		
1c	What are polymers? Mention two common applications.	2		
1d	Differentiate between an open system and a closed system in context of thermodynamics.	2		
1e	Define casting. Why is it used in industry?	2		
2a	A rod AB , hinged at A and attached to cable BD at the point B , supports the loads shown as shown in the figure below. Knowing that $d = 200$ mm, determine (a) the tension in cable BD , (b) the reaction at A .			

NME 101 | 2025-26 | Q1(a)-(e) Mechanism vs machine, zeroth & third law, polymers, open/closed system, casting